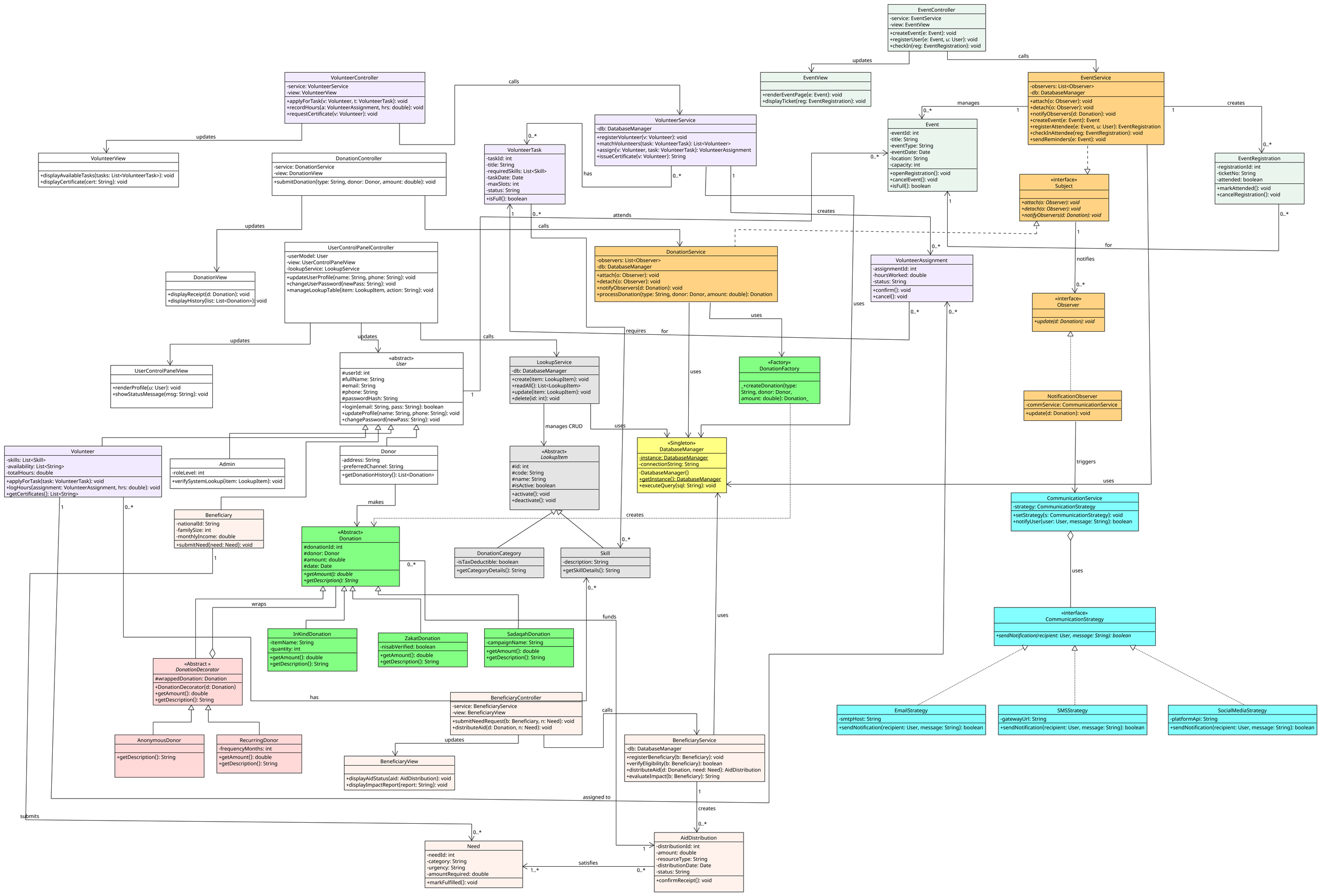
EventController
-service: EventService
-view: EventView
+createEvent(e: Event): void
+registerUser(e: Event, u: User): void
+checkIn(reg: EventRegistration): void
updates
calls
EventView
+renderEventPage(e: Event): void
+displayTicket(reg: EventRegistration): void
EventService
-observers: List<Observer>
-db: DatabaseManager
+attach(o: Observer): void
+detach(o: Observer): void
+notifyObservers(d: Donation): void
+createEvent(e: Event): Event
+registerAttendee(e: Event, u: User): EventRegistration
+checkInAttendee(reg: EventRegistration): void
+sendReminders(e: Event): void
manages
1
0..*
creates
Event
-eventId: int
-title: String
-eventType: String
-eventDate: Date
-location: String
-capacity: int
+openRegistration(): void
+cancelEvent(): void
+isFull(): boolean
«interface»
Subject
+attach(o: Observer): void
+detach(o: Observer): void
+notifyObservers(d: Donation): void
EventRegistration
-registrationId: int
-ticketNo: String
-attended: boolean
+markAttended(): void
+cancelRegistration(): void
for
notifies
«interface»
Observer
+update(d: Donation): void
NotificationObserver
-commService: CommunicationService
+update(d: Donation): void
triggers
uses
CommunicationService
-strategy: CommunicationStrategy
+setStrategy(s: CommunicationStrategy): void
+notifyUser(user: User, message: String): boolean
«interface»
CommunicationStrategy
+sendNotification(recipient: User, message: String): boolean
EmailStrategy
-smtpHost: String
+sendNotification(recipient: User, message: String): boolean
SMSStrategy
-gatewayUrl: String
+sendNotification(recipient: User, message: String): boolean
SocialMediaStrategy
-platformApi: String
+sendNotification(recipient: User, message: String): boolean
VolunteerController
-service: VolunteerService
-view: VolunteerView
+applyForTask(v: Volunteer, t: VolunteerTask): void
+recordHours(a: VolunteerAssignment, hrs: double): void
+requestCertificate(v: Volunteer): void
VolunteerView
+displayAvailableTasks(tasks: List<VolunteerTask>): void
+displayCertificate(cert: String): void
VolunteerService
-db: DatabaseManager
+registerVolunteer(v: Volunteer): void
+matchVolunteers(task: VolunteerTask): List<Volunteer>
+assign(v: Volunteer, task: VolunteerTask): VolunteerAssignment
+issueCertificate(v: Volunteer): String
VolunteerTask
-taskId: int
-title: String
-requiredSkills: List<Skill>
-taskDate: Date
-maxSlots: int
-status: String
+isFull(): boolean
has
attends
VolunteerAssignment
-assignmentId: int
-hoursWorked: double
-status: String
+confirm(): void
+cancel(): void
DonationController
-service: DonationService
-view: DonationView
+submitDonation(type: String, donor: Donor, amount: double): void
DonationView
+displayReceipt(d: Donation): void
+displayHistory(list: List<Donation>): void
UserControlPanelController
-userModel: User
-view: UserControlPanelView
-lookupService: LookupService
+updateUserProfile(name: String, phone: String): void
+changeUserPassword(newPass: String): void
+manageLookupTable(item: LookupItem, action: String): void
UserControlPanelView
+renderProfile(u: User): void
+showStatusMessage(msg: String): void
DonationService
-observers: List<Observer>
-db: DatabaseManager
+attach(o: Observer): void
+detach(o: Observer): void
+notifyObservers(d: Donation): void
+processDonation(type: String, donor: Donor, amount: double): Donation
requires
for
uses
«abstract»
User
#userId: int
#fullName: String
#email: String
#phone: String
#passwordHash: String
+login(email: String, pass: String): boolean
+updateProfile(name: String, phone: String): void
+changePassword(newPass: String): void
LookupService
-db: DatabaseManager
+create(item: LookupItem): void
+readAll(): List<LookupItem>
+update(item: LookupItem): void
+delete(id: int): void
manages CRUD
«Factory»
DonationFactory
_+createDonation(type:
String, donor: Donor,
amount: double): Donation_
«Singleton»
DatabaseManager
-instance: DatabaseManager
-connectionString: String
-DatabaseManager()
+getInstance(): DatabaseManager
+executeQuery(sql: String): void
«Abstract»
LookupItem
#id: int
#code: String
#name: String
#isActive: boolean
+activate(): void
+deactivate(): void
DonationCategory
-isTaxDeductible: boolean
+getCategoryDetails(): String
Skill
-description: String
+getSkillDetails(): String
Volunteer
-skills: List<Skill>
-availability: List<String>
-totalHours: double
+applyForTask(task: VolunteerTask): void
+logHours(assignment: VolunteerAssignment, hrs: double): void
+getCertificates(): List<String>
Admin
-roleLevel: int
+verifySystemLookup(item: LookupItem): void
Donor
-address: String
-preferredChannel: String
+getDonationHistory(): List<Donation>
Beneficiary
-nationalId: String
-familySize: int
-monthlyIncome: double
+submitNeed(need: Need): void
makes
«Abstract»
Donation
#donationId: int
#donor: Donor
#amount: double
#date: Date
+getAmount(): double
+getDescription(): String
wraps
funds
InKindDonation
-itemName: String
-quantity: int
+getAmount(): double
+getDescription(): String
ZakatDonation
-nisabVerified: boolean
+getAmount(): double
+getDescription(): String
SadaqahDonation
-campaignName: String
+getAmount(): double
+getDescription(): String
«Abstract »
DonationDecorator
#wrappedDonation: Donation
+DonationDecorator(d: Donation)
+getAmount(): double
+getDescription(): String
AnonymousDonor
+getDescription(): String
RecurringDonor
-frequencyMonths: int
+getAmount(): double
+getDescription(): String
has
BeneficiaryController
-service: BeneficiaryService
-view: BeneficiaryView
+submitNeedRequest(b: Beneficiary, n: Need): void
+distributeAid(d: Donation, n: Need): void
BeneficiaryView
+displayAidStatus(aid: AidDistribution): void
+displayImpactReport(report: String): void
BeneficiaryService
-db: DatabaseManager
+registerBeneficiary(b: Beneficiary): void
+verifyEligibility(b: Beneficiary): boolean
+distributeAid(d: Donation, need: Need): AidDistribution
+evaluateImpact(b: Beneficiary): String
assigned to
submits
Need
-needId: int
-category: String
-urgency: String
-amountRequired: double
+markFulfilled(): void
AidDistribution
-distributionId: int
-amount: double
-resourceType: String
-distributionDate: Date
-status: String
+confirmReceipt(): void
satisfies